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Intelli-Mirror: An Augmented Reality based IoT System for Clothing and Accessory Display

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***Abstract*— The paper presents an Augmented Reality (AR) based implementation of an interactive system termed "Intelli-Mirror". The Intelli-Mirror uses Image Processing techniques to detect a user and then displays a garment image on the person. The Intelli-Mirror provides the user with the ability to switch between the selections available in an entire inventory of clothing. This negates the need of continuously having to physically change clothing every time a customer wishes to try on a new garment. This effectively reduces the time to shop to just the decision-making time. The implementation thus provides relief from many of the problems associated with physically changing clothing. The Intelli-Mirror is also an Internet of Things (IoT) device implemented on a Raspberry Pi capable of updating itself to the newest inventory. It can also provide data for user analytics to aid in clothing design and marketing.**

***Keywords*— Augmented Reality, Internet of Things, Computer Vision, Image Processing, Raspberry Pi, User Analytics**

I. INTRODUCTION

Given the fast pace of the modern world, there is a need to optimize the current system in traditional retail stores, wherein customers who wish to view themselves in different clothes must physically change outfits. This requirement is very time-consuming, and comparatively, the customers spend a small amount of time actually viewing themselves in the clothes, and their choice of outfits is limited to as many as are possible to carry into a trial room. This can create problems and is a highly inefficient system.

Retailers risk damaging the outfits as customers put on and take off the clothes. Customers may not be permitted to, or would perhaps hesitate to try on an expensive item for fear of possible wear and tear, thus consequently reducing the chances of a purchase as well. Again, certain items that are in storage cannot be shown to the customer, because they have not been made display ready, may have just arrived in the warehouse or may still not have been dispatched. Additionally, there is also transference of the previous user's state of hygiene onto the items of clothing. This is a disconcerting factor for many customers who would prefer not to try on such items and prefer a new piece that is still in original packing. This could lead to wastage. Likewise, at home, a person may have a substantial number of items of clothing but only use a select few because they are personally preferred or merely accessible. This results in an inefficient use of storage space and a lot of clothing items are bought, and never used.

The Intelli-Mirror is the solution for such problems. It is an Augmented Reality system that displays clothing items on users. A representation of the system is shown in Fig. 1.

Fig. 1. Graphical Representation of the Intelli-Mirror

The implementation is carried out on a Raspberry Pi. The device updates itself with the current inventory, and can return user analytics data to a central server. The implementation is based in Python using the OpenCV[1] and Numpy[2] Libraries allowing direct portability to the Raspberry Pi.

A. *Proposed Implementation*

At boot, the system checks with a server about changes in the inventory, and updates itself with the corresponding images and their attributes. The system then scans for users. If a user stands in front of the Intelli-Mirror for more than 5 seconds, the system starts up, and the user is asked to select a store or the kind of clothing they wish to view. The user is then asked to define their marker thresholds. A marker is any recognizable object the user wants to track, ie. a shirt, or a colored object. The image of the user is checked for markers and if detected satisfactorily, the corresponding images are displayed over the user's body. Information about the clothing selection such as Name, Price, UID, and Size are also displayed. The user can then toggle between the items available in the inventory. The user can choose to have certain items added to their cart. Upon exit, the system displays the final cart and the corresponding bill. This ends a single user transaction and the Intelli-Mirror reverts back to a simple mirroring display, and awaits the next user interaction. The basic block diagram of the system is shown in Fig. 2.

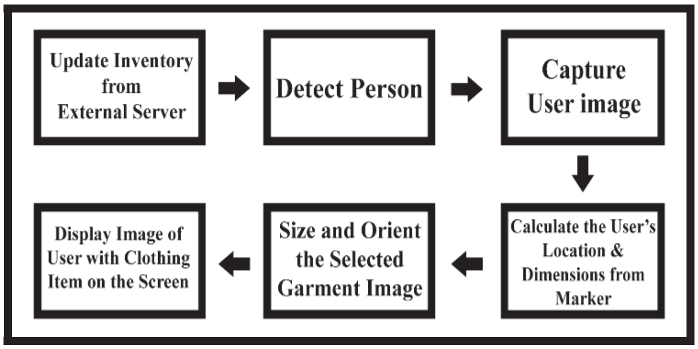

Fig. 2. Basic Block Diagram of the Intelli-Mirror

II. AUTO-START

A. Introduction

The Intelli-Mirror is self-starting, that is, if a person enters the range of view of the camera, and remains for 5 seconds, the Intelli-Mirror system starts up. The display image switches from a simple Mirror view to the start screen. If the user does not interact with the system for 1 minute, the view switches back to the simple Mirror view, where the display is merely reproducing the camera output. This implementation assumes the Intelli-Mirror is grounded and will not be subject to movement, but the 1 minute timeout allows for any disturbances in this respect. The video output from the camera of the Intelli-Mirror can also be used as a security camera feed, adding extra functionality. The system described below differs from using a basic approach sensor or radar sensor, similar to the kind used in automatic swing doors, in that it will only respond to the intrusion of a full being in a stationary position, and not merely a person passing by.

B. Implementation

The initial image, I_R, is captured and it is held as a reference image. As new frames, I_n, are captured, they are compared with the reference image as shown in Fig. 3. This is done by finding the difference, I_K, between the 2 images as in [1].

$$I_K = |(I_R - I_n)| \tag{1}$$

If there is an addition to the reference image, in terms of a person moving into the frame, as shown in Fig. 3, a timer begins.

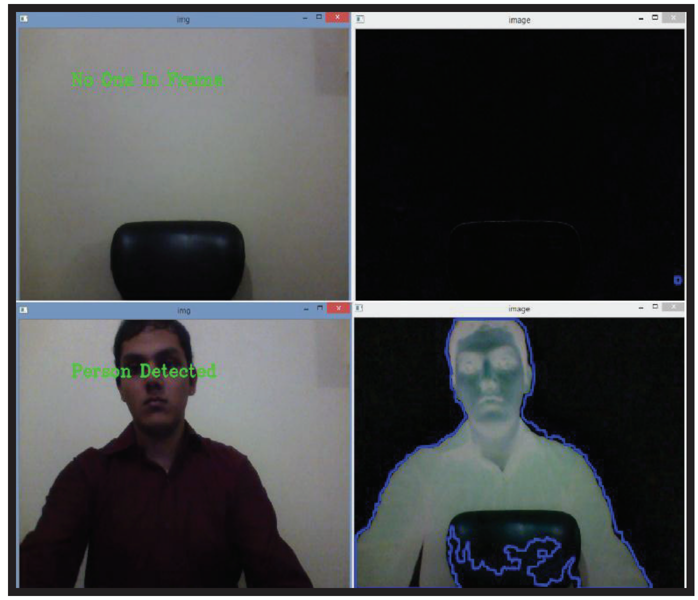

Fig. 3. Top: Absence of a person in frame.
Bottom: Presence of a person is detected.

If the timer exceeds 5 seconds, this is recognized as a user being present in the frame and the display switches to the start screen. If the no user input is received in 1 minute, the display switches back to a plain mirror style view and the process is repeated.

If a user is not detected for 10 seconds, the reference image, I_R, is changed to the current camera frame to account for any disturbances or shift of frame affecting detection.

III. IMAGE PROCESSING

A. Marker Detection

Color Markers are used for detection as color is an image parameter that does not suffer much from distortion or changes in orientation and location. The camera input is captured as a BGR (Blue, Green, and Red) matrix having values from 0 to 255 for each of the respective color channels.

This image is then filtered with a median filter to remove extreme-valued noise. The median filter takes a section of the image (here, a 3x3 matrix) and orders every pixel (3x3= 9 pixels) based on its value. The median, or halfway number of this selection is taken and the center pixel of the 3x3 matrix is replaced by the median value. This removes any values that are too high, or too low from affecting the image. These extreme values are thus filtered out of the image with no disturbance to the edges.

Next, the image is converted to HSV (Hue, Saturation, and Value) effectively separating the chrominance, or color information, from the luma, or image intensity, allowing for easier thresholding. This is akin to a conversion from a Cartesian co-ordinate system to a Cylindrical co-ordinate system, where the radial distance indicates color. This step is critical because it converts the color information from being spread out among the Blue, Green and Red channels, into a single Hue channel. The Equations [2, 3, 4] describe BGR to HSV conversion.

$$V = \max(R,G,B) \tag{2}$$

$$H = \begin{cases} 60\,(G - B)/(V - \min(R, G, B)) & \text{if } V = R \\ 120 + 60\,(B - R)/(V - \min(R, G, B)) & \text{if } V = G \\ 240 + 60\,(R - G)/(V - \min(R, G, B)) & \text{if } V = G \end{cases} \tag{3}$$

$$S = \begin{cases} (V - \min(R, G, B))/V & \text{if } V \neq 0 \\ 0 & \text{if } V = 0 \end{cases} \tag{4}$$

Following this, the image, f(x, y), is thresholded based on values of HSV, with the output, g(x, y) retaining only a binary version of the image as in [5],

$$g(x, y) = \begin{cases} 1 & \text{if } T_1 < f(x, y) < T_2 \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

where, T_1 is the lower threshold and T_2 is the upper threshold value.

The thresholded image is as seen in Fig. 4.

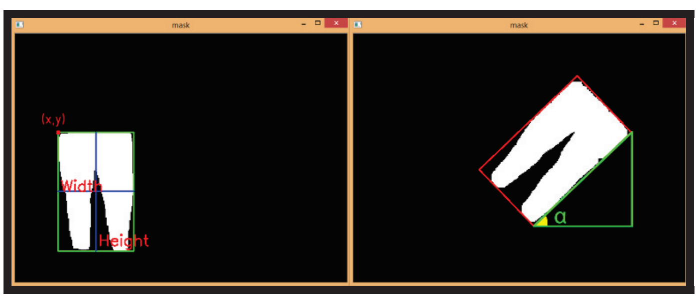

Fig. 4. The white region indicates pixels within the threshold. The Marker is detected and its Location, Dimension and Orientation are calculated.

Finally, the image is searched for contours that represent the boundaries of the shapes in the image. If a suitable contour is detected, the contour is taken to be the desired marker.

The area, position and orientation parameters of the marker are calculated from image moments and supplementary equations[4]. For an image, f(x, y), of size M x N, the 2-D moment of order (p + q) is given by [6],

$$m_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} x^p\, y^q f(x, y) \tag{6}$$

where p = 0, 1, 2,… and q = 0, 1, 2,… $\in \mathbb{Z}$.

The central moment of order (p + q) is defined as in [7],

$$\mu_{pq} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} (x - \bar{x})^p (y - \bar{y})^q f(x, y) \tag{7}$$

where,

$\bar{x} = \dfrac{m_{10}}{m_{00}}$ and $\bar{y} = \dfrac{m_{01}}{m_{00}}$.

The normalized central moments, denoted as η_{pq}, are defined in [8],

$$\eta_{pq} = \frac{\mu_{pq}}{\mu_{00}^{\gamma}} \tag{8}$$

The moments give a description of the marker which is used to size the clothing image. Special care must be taken to accommodate the orientation (θ) of the marker as calculated from [9],

$$\theta = \frac{1}{2} \tan^{-1}\left[\frac{2\mu(1,1)}{\mu(2,0) - \mu(0,2)}\right] \tag{9}$$

.

B. *Ad-hoc Marker Selection*

The Intelli-Mirror has a system that allows the user to use a mouse click-and-drag operation to select their desired color marker. This could also be implemented on a touch screen display. The HSV values inside the contour region are calculated, and these are used to describe the corresponding marker is shown to the user as in Fig. 5.

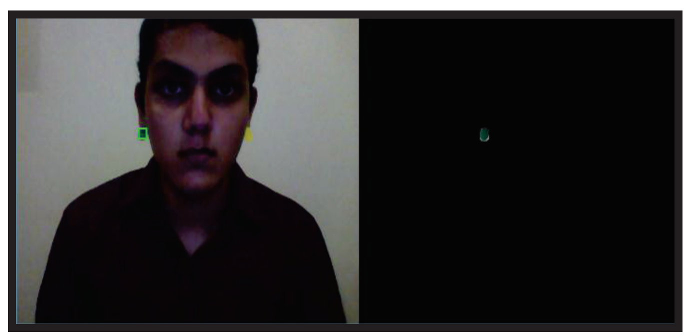

Fig. 5. Left: Green Marker as selected (in green) by user is detected and the result displayed to the user.

This avoids the issue of having to either constantly input threshold values or relying on only a specific set of values. If the user is satisfied, they may move on to the next step, or else redraw the marker boundary till it is acceptable. The camera capture that is used to find the marker can be refreshed by the user at will.

C. Clothing Image-Specific Transformations and Results

1) Single Marker Transformations

First, the location of the vertices of the marker are determined and the height and width of the markers are calculated from it. The angle with the horizontal is also calculated by trigonometric functions as described in the Fig. 4.

The image of the respective clothing item is loaded. This image is then warped based on the marker dimensions and placed on top of the marker. This creates an image of the user wearing the garment as in Fig. 6.

Fig. 6. The image of the shirt is displayed on the user's frame.

Because the image is warped precisely to the marker, described also by its angle, the user may lean or move at any inclination without disturbance in the location of the image as in Fig. 7.

Fig. 7. The image of the trouser tilts with the user.

2) Double Marker Transformations

For the display of necklaces, spectacles, earrings and ties, two markers are detected and they are treated as a single marker with respect to orientation and location. This combined marker is then used to size the images of the respective clothing items.

The physical marker here is constructed of two colored plastic objects, connected by a flexible wire. This may be 3D Printed.

For necklaces, neckties and bow ties, the images are sized based on distance between the two markers. This represents the width of the user's neck. The height of the clothing image is calculated with respect to the marker width.

For spectacles, the distance between the markers represents the width of the face, from temple to temple. The image of the spectacles is then placed within this space.

If the user turns their face to the right or left, the distance between the markers also changes, as would the temple to temple width of the face change in a 2D image. This causes the spectacles' image to shrink in width giving it a natural appearance. The results for Necklaces, Spectacles and Ties are as shown in Fig. 8.

Fig. 8. Top Left: Gold Chain, Top Right: Spectacles, Bottom Left: Bow Tie, Bottom Right: Tie.

For earring displays, a earring is drawn on each of the visible markers.

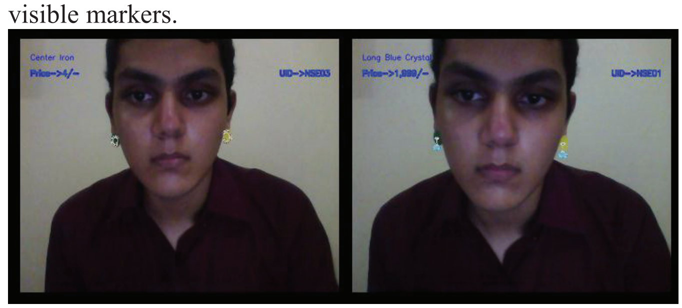

Fig. 9. Left: Vertically Flipped Earrings. Right: Dangling Earrings.

If both markers are detected, the earring to the left is flipped along the vertical axis if necessary to represent a set of earrings facing each other as in Fig. 9. If the images of the earrings are of those that are attached to a small chain (this must be defined in the image properties), the earring image is placed at a lower height to bring the hook at level with the ear as in the right-side image of Fig. 9.

IV. RASPBERRY PI AND THE INTERNET OF THINGS (IOT)

The hardware requirements for the Intelli-Mirror are a display monitor, a good quality camera having a capture rate of at least 15-20 frames per second, an input device like a mouse, an Internet connection and a suitably fast microprocessor that can interface with the aforementioned items. The Raspberry Pi Board is a device that fits such requirements. The Intelli-Mirror can also be implemented on a Thin Client.

The Raspberry Pi model 2 B as shown in Fig. 10. is used in the implementation.

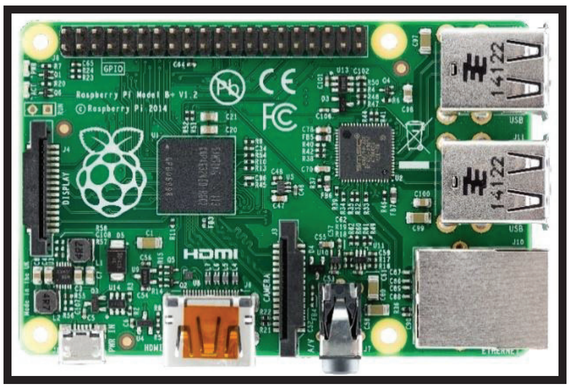

Fig. 10. Raspberry Pi Model 2B

The Intelli-Mirror is connected to the Internet via a Wi-Fi Dongle plugged into the Raspberry Pi. The Raspberry Pi Model 2B is a single board computer with 1GB of RAM, and a 900 MHz quad-core ARM Cortex-A7 processor in a Broadcom BCM2836 SoC (System on Chip). It has USB interfaces for input devices and can be connected to the Internet via Ethernet or a Wi-Fi Module. Its video output can directly be displayed on an HDMI screen.

Each Intelli-Mirror has its own device ID. This serves as an easy fault detection service and helps keep track of the data it transmits. In the implementation, the Raspberry Pi pings the server after boot, checking for any new updates to the inventory. This update can also be interrupt based, with the Raspberry Pi checking if the server has updated in a time-bound manner, for example, every two hours. All missing elements are downloaded to the media cache, and items no longer available for sale are deleted. Corresponding data such as Name, Price, Store of Origin, Sizing, and Type of Clothing are also copied from the server, and indexed accordingly. The Raspberry Pi then uploads its user analytics data to the server, and may discard its own copy in favor of a model that has been generated by the server. Thus all Intelli-Mirrors is connected to each other via the server, thereby sharing information.

V. CONSUMER-ORIENTED DESIGN

A. Graphical User Interface (GUI)

Users can select if they wish to view the catalogue of the entire store, a specific brand or a specific type of clothing. Based on the Selection, the respective products are available for display. Each inventory item has attributes such as the Name, Price and Unique ID (UID). These are displayed along with the image on the Intelli-Mirror.

B. Fiducial Markers

A system of Fiducial Markers is used to size clothing based on international standards. Fiducials are items in the camera's view whose size is known and which can be used for distance approximations. In the implementation, a set of markers is used, the distance between which is fixed. This provides an estimate for the user's height, and width of shoulders or waist. This distance can be used to find the best fitting size that the user should purchase. Typical shirt width to length sizes are in the ratio of 1:1.47[11]. This ratio can be used to approximate the user's size as well for user analytics.

C. Billing System

Any items that the user selects to purchase are added to the cart. The user is informed of the amount of items purchased and the cost and can add to or remove from the list of items purchased at any point. When the user choses to end the transaction, the bill is produced. This may be transmitted to them if they are connected to the Intelli-Mirror. A sample bill is as shown in Fig. 11.

```
Your Final Cart=
NSG01 :  4 x 123 = 492
NSN01 :  1 x 645 = 645
NSN02 :  1 x 1999 = 1999
NSN03 :  1 x 1000 = 1000
NSG04 :  1 x 4300 = 4300
NSG06 :  2 x 5005 = 10010
NST01 :  2 x 5007 = 10014
NSP03 :  3 x 50448 = 151344
NSP01 :  2 x 210 = 420
NSE03 :  1 x 4315 = 4315
NSE02 :  1 x 2999 = 2999
NSE01 :  1 x 504 = 504
NSS01 :  2 x 10075 = 20150

Your Bill= 1,82,949/-
Thank you for using the Intelli-Mirror
```

Fig. 11. Bill Display

VI. FUTURE SCOPE AND APPLICATIONS

A. User Analytics

The Intelli-Mirror may be used to generate data about the patterns of customer preferences. Information such as typical dimensions of users may be collected. The device can also detect the time duration spent by a user on a specific product. If, for example, users spend a lot of time viewing a specific type of clothing, but do not purchase it, the design team can be alerted to the fact that users show an interest in the product but are unwilling to complete purchase due to a discouraging factor such as its price or color.

Trends can be spotted from the type of clothing styles that are favored by the user. Also, the common sizes of typical users can be calculated and accordingly designers can produce more products that specific size, allowing for more efficient sales. Such information could be used by the marketing department to devise new strategies and develop products according to what users tend to buy.

B. Social Media Connect

The system could connect to a user's mobile device, and display clothing choices based on their previous selections or preferences. Users could then share information about their purchases, and even UIDs for other users to purchase similar items. Items that the users add to cart can be transmitted to their phone along with relevant information such as price and size creating a virtual billing system. The user could take a screenshot of the camera's current display and have the image sent to a mobile device as a souvenir or in order to make a decision to purchase at a later date.

C. Intelli-Mirror for the Home User

The Intelli-Mirror can be used by a home users as well, allowing for an optimized storage system. A home user could have their entire wardrobe implemented in the system. This would allow for stacking of clothes in a space optimized manner without the need for clothes to be arranged for easy access as the system would know the location of every clothing item. It would also reduce time spent searching for a selected clothing item as the user has to merely make their choice and a robotic arm would move to the respective location and retrieve the clothing item. Worn clothes could be cleaned and stored away without the need of human effort wherein a system would perform all tasks of washing, drying, folding and storing autonomously.

The system could also allow for multiple user inventories, where the user could be detected based on their body shape, or have a pin based security system. The respective users clothing inventory would be selected and displayed. The robotic system would then access only their specific clothing items.

Users could select certain choices as favorites, and these items would be displayed at the start of the list based on priority. Alternatively, users could set the system to avoid repeating a wardrobe option or notify the user if an item was worn recently or how often it has been worn.

VII. CONCLUSION

The current implementation of the Intelli-Mirror system covers the display of Shirts, Trousers, Necklaces, Earrings, Spectacles and Ties. It is implementable in clothing stores where the display space of the store can be optimized to include some areas for the Intelli-Mirror system. The system thus provides an innovative new platform for designers to showcase their products.

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The current article is a continuation of research in the above reference by the author. The current submission contains extensions into double marker based detection, an ad-hoc marker selection system, item attribute access for user analysis, self-starting by background subtraction and details an IOT based implementation termed "Intelli-Mirror".

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